

Tanisha Bipin Shrotriya | [Email](#) | [Phone](#) | [GitHub](#) | [LinkedIn](#)

Summary

Software engineer with 6 years of experience building and stabilizing large-scale backend and data systems. I focus on making complex, high-stakes systems reliable by improving data integrity, reducing operational overhead, and enabling better decision-making. Known for owning ambiguous problems, aligning across teams, and delivering practical, high-impact solutions.

Skills C#, Java, Python, Postgres, Azure Data Factory, Docker, Azure SQL, PowerBI, Scala, ServiceNow, Latex, MySQL, Cassandra, DynamoDB, PyTorch, Pandas, Numpy, CUDA, TensorFlow, C++, C, R, JavaScript, Node.js, Linux, JIRA, Django, Flask

Experience

Software Engineer *Microsoft Corporation – OneDrive SharePoint* — 2023 – 2025, (Intern 2022)

- Eliminated **100%** legal exposure for Microsoft, by leading the integration testing, allowing the feature to be turned on for all consumer One Drive accounts migrated to SharePoint Online stack. (Sandbox Testing)
- Reduced on-call effort from **1 hour to 10 minutes** in **2 days** by building automation with recovery flows, fallbacks, and clear operational runbooks. (Shell Scripting)
- Owned design and deployment of the OneDrive Largest Files API, restoring a broken feature and scaling it to **~250M** users and **~30M** daily requests within **2 months**. (C#, Unit Testing)
- Designed and deployed **GDPR-compliant** anomaly detection pipelines (hourly/daily) for Digital Safety systems, generating proactive alerts that **prevented** scaling-related production incidents and **eliminated** recurring security and privacy audit penalties. (Azure Data Factory, Azure SQL, PowerBI)
- Improved telemetry coverage impacting data collection stage of these pipelines, by **45%** in **2 weeks** by collaborating with partners owning data ingestion streams. (C#)
- Identified and resolved a major data duplication issue and uncovered a hidden data loss bug, improving metric accuracy by **50%** across 2024–2025 data for the Enterprise Digital Safety Pipelines
- Designed guardrails and alerts for incorrect data access by law enforcement agents, reducing noisy Enterprise Digital Safety pipeline metrics by **15%** and improving privacy compliance.
- Engaged in **50+** code reviews with quality improvement suggestions for code written by team setting standards during early adoption stages of copilot across 4 business domains by mentoring juniors and holding seniors accountable. (C#, SQL)

Research Assistant *Knowledge Computing Lab, UMN* — 2023

- **System Integration Project** – Built a [Dockerized](#) full-stack ML workflow integrating [MapKurator \(GAN\)](#) with [Recogito](#) for large-scale historical map processing, automating label extraction and reducing manual labeling to validation; supported adoption via YouTube [tutorials](#) (**430+** views, **608** [downloads](#)). (Python, Scala)
- **DARPA Critical Mineral Extraction Challenge** – Combined exploratory data analysis with stakeholder collaboration to clarify complex input data requirements, simplify scope, and align ML system design, accelerating execution and contributing to a **1st Prize** finish. [Poster](#) (Python)

Software Engineer – Service Operations *American Express India Pvt. Ltd.* — 2019 to 2021

- Designed an automated incident resolution system to address recurring issues decreasing toil **by 70%** that leveraged data from ServiceNow and internal AI-driven NLP API plug-in as a stretch assignment. (NodeJS)
- Resolved sev2 incidents over 2 years for 7 merchant applications, ensuring service continuity without SLA breach or escalation to sev1. (ServiceNow, Informatica, Kafka, Linux Servers)

Education:

- **University of Minnesota, Twin Cities** Master's in Computer Science, Minor Data Science – GPA: 3.5

Projects

- **Attention-Based Damage Detection (xView2):** Implemented Siamese CNN with attention achieving 98% recall for disaster response based on [Hao et al](#) for [xView2](#) Dataset. (Other Metrics: Accuracy: **71%**, Precision: **26%**) ([Paper](#)) ([GitHub](#)) (AI/ML/DataScience: Python, PyTorch, Pandas, Numpy)
- **Transfer Learning Driven Personalized Recommendation Web Application:** Accuracy **95%** Built an MCP-style full-stack workflow applying Transfer Learning with MobileNet to deliver furniture recommendation. ([GitHub](#)) (AI/ML/DataScience/FullStack: Python, Tensorflow, Keras, Pandas, Numpy, Django, MySQL, Data processing)
- **LSTM Stock Forecasting:** Designed RNN model outperforming ARIMA by **25%** lower RMSE for long-term prediction. ([Portfolio Optimization GitHub Link](#)) (AI/ML/DataScience: Python, TensorFlow, Numpy, Pandas, R)
- **Drought Prediction System (Smart India Hackathon):** Led a 6-member team developing ANN-based environmental time-series predictor (**75% accuracy**). (AI/ML/DataScience: Python, TensorFlow, Keras, Excel, Flask, HTML, CSS)
- **Medicare Database Analysis:** Evaluated scalable data stores for analytical workloads, identifying **Cassandra** for bulk-load scalability, **DynamoDB** limitations for high-throughput analytics, and **Snowflake** as the optimal solution combining SQL flexibility with large-scale performance. ([Paper Link](#)) (Big Data Engineering)
- **Distributed File Sharing Application:** Developed a distributed file-sharing desktop app using Java RMI, implementing multi-user CRUD operations with consistency checks across networked systems. (Distributed Systems, Networking, Backend, OOP)

Publications

[Acknowledged Contributor](#), “The mapKurator System: A Complete Pipeline for Extracting and Linking Text from Historical Maps,” ACM SIGSPATIAL, 2023

[Poster Presenter](#), “Multiclass Classification of Global Water Bodies in ReaLSAT,” American Association of Geographers Conference, 2023 applied transfer learning and clustering for large-scale remote sensing waterbody.

[Technical Author XRDS: The ACM Student Magazine](#): Articles based on sessions attended at the ACM Natural Language Processing Summer School, IIT Kharagpur (2017).

Achievements

- 1st Prize, Team ISI-UMN for Critical Mineral Extraction Challenge 2022
- 3rd Prize HackTheGap 2021
- Team Leader Smart India Hackathon 2018
- Attendee, Natural Language Processing Summer School, IIT Kharagpur 2017
- Research Fellow, Summer School in Math & Finance, Chennai Mathematical Inst. 2018
- CCOEW ACM-W Student Chapter Chair 2017 - 2019
- WeTech IIE Goldman Sachs Women in Tech Scholar 2017 - 2018
- CCOEW Robotics Coding Core Team 2015 – 2017

Volunteer Experience

Code Path Volunteer: Mentoring, Resume Reviews and Career Guidance for Code Path students.

Tilth Alliance, Seattle: Master Composter/ Sustainability Steward for the city of Seattle

PAWS, Snohomish: Seasonal WildLife Care Assistant 2026